MTH636/SDS636 (2026-27, ODD SEMESTER)

PRACTICE PROBLEM SET 2

1. There are n individuals who witness a crime. Everybody would like the police to be called. If this happens, each individual derives satisfaction $v > 0$ from it. Calling the police has a cost of c, where $0 < c < v$. The police will come if at least one person calls. Hence, this is an n-person game in which each player chooses from $\{C, N\}$; C means 'call the police' and N means 'do not call the police'. The payoff to person i is 0 if nobody calls the police, $v - c$ if i (and perhaps others) call the police, and v if the police is called but not by person i.

   (a) What are the Nash equilibria of this game in pure strategies? Does the game have a symmetric Nash equilibrium in pure strategies (a Nash equilibrium is symmetric if every player plays the same strategy)?

   (b) Compute the symmetric Nash equilibrium or equilibria in mixed strategies.

   (c) For the Nash equilibrium/equilibria in (b), compute the probability of the crime being reported. What happens to this probability if n becomes large?

2. There are two cities S and T. A total of 4000 cars need to go from S to T. There are two paths from S to T. Path 1 goes via city A — we will denote this as $S \to A \to T$. Path 2 goes via city B — we will denote this as $S \to B \to T$. Note that roads go in one direction.

   On Path 1, the road from S to A takes $\frac{x}{100}$ units of time, where x is the number of cars traveling on this road. On the other hand, the road from A to T takes 45 units of time, irrespective of the number of cars traveling on this road. On Path 2, the road from S to B takes 45 units of time, irrespective of the number of cars traveling on this road. On the other hand, the road from B to T takes $\frac{y}{100}$ units of time, where y is the number of cars traveling on this road.

Suppose the utility of every car is the negative of the amount of time spent in going from S to T.

(a) Suppose each car (driver) is strategic in choosing his path and all cars choose their paths simultaneously. Describe a pure strategy Nash equilibrium of this game.

The planner decided to build another road to travel from A to B (you cannot travel from B to A on this road). The travel time from A to B is zero (irrespective of the number of cars traveling on this road). So, now there are three paths from S to T: Path 1 is $S \to A \to T$, Path 2 is $S \to B \to T$, and Path 3 is $S \to A \to B \to T$.

(b) Suppose each car (driver) is strategic in choosing his path and all cars choose their paths simultaneously. Does the Nash equilibrium that you found when the road from A to B did not exist still a Nash equilibrium now? Describe a pure strategy Nash equilibrium of this game.

(c) What conclusions can be drawn from this Nash equilibrium about whether the planner should build the road from A to B?

3. Consider a game $G = \langle N = \{1,2\}\}, (S_i)_{\{i \in N\}}, (u_i)_{\{i \in N\}} \rangle$ where $|S_i| = 2$ for $i \in \{1,2\}$. Prove or disprove the following statements.

(a) Suppose G has exactly one pure strategy Nash Equilibrium. There cannot be any other Nash Equilibrium in mixed strategies.

(b) Suppose we have the following two pure strategy equilibria for G (recall that one such game (battle of the sexes) we discussed in class).

	s_1^2	s_2^2
s_1^1	NE	Not NE
s_2^1	Not NE	NE

There is a another equilibrium $\sigma^\star$, different from the two pure strategy equilibria, in the game.

(c) Suppose no player has a strictly dominated strategy in G. There can be infinitely many equilibria in G.

4. Consider a game $G = \langle N, (S_i)_{\{i \in N\}}, (u_i)_{\{i \in N\}} \rangle$ where S_i is finite for all $i \in N$. Suppose the players are pessimistic and the utility of player i for a mixed strategy profile σ is defined as follows:

$$\hat{u}_i(\sigma) = \min\{u_i(s) \mid s \in \text{Supp}(\sigma)\}$$

where $\text{Supp}(\sigma) = \{(s_1, \ldots, s_n) \in S_1 \times \cdots \times S_n \mid \sigma_i(s_i) > 0 \text{ for all } i \in N\}$. Either prove or disprove the following: The game G has a mixed strategy NE when players are pessimistic.

5. Suppose that a mixed strategy σ_i of player i strictly dominates another of his mixed strategies, $\hat{\sigma}_i$. Prove or disprove each of the following claims:

   (a) Player i has a pure strategy $s_i \in S_i$ satisfying: (i) $\hat{\sigma}_i(s_i) > 0$ and (ii) strategy s_i is not chosen by player i in any equilibrium.

   (b) For each equilibrium $\sigma^* = (\sigma_i^*)_{\{i \in N\}}$, player i has a pure strategy $s_i \in S_i$ satisfying (i) $\hat{\sigma}_i(s_i) > 0$ and (ii) $\sigma_i^*(s_i) = 0$.

6. Let $(a_{i,j})_{1 \leq i,j \leq n}$ be non-negative numbers satisfying $\sum_{j \neq i} a_{i,j} = a_{i,i}$ for all $i \in \{1, \ldots, n\}$. Julie and Sam are playing the following game. Julie writes down a natural number i, $1 \leq i \leq n$, on a slip of paper. Sam does not see the number that Julie has written. Sam then guesses what number Julie has chosen, and writes his guess, which is a natural number j, $1 \leq i \leq n$, on a slip of paper. The two players simultaneously show each other the numbers they have written down. If Sam has guessed correctly, Julie pays him $a_{i,i}$ dollars, where i is the number that Julie chose (and that Sam correctly guesses). If Sam was wrong in his guess ($i \neq j$), Sam pays Julie $a_{i,j}$ dollars. Depict this game as a two-player zero-sum game in strategic form, and prove that the value in mixed strategies of the game is 0.